TARNDEEP SINGH

AI & Machine Learning | B.Tech CSE (Data Science), 2027

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PROFESSIONAL SUMMARY

B.Tech Computer Science and Engineering student specialising in Data Science at Akal University (CGPA: 9.02/10), focused on machine learning, deep learning, and applied AI development. Hands-on experience building sequence-based recognition systems, molecular classification pipelines, and retrieval-grounded AI applications. Skilled in Python, PyTorch, TensorFlow, Scikit-learn, computer vision, feature engineering, model evaluation, and deploying inference services.

TECHNICAL SKILLS

Programming: Python, Java, SQL, JavaScript

Machine Learning: Supervised Learning, Classification, Multi-label Classification, Feature Engineering, Data Preprocessing, Data Augmentation, Cross-Validation, Hyperparameter Tuning, Model Evaluation

Deep Learning and AI: Neural Networks, Bi-LSTM, Attention Mechanisms, Sequence Modelling, Computer Vision, Retrieval-Augmented Applications

Libraries and Frameworks: PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, OpenCV, MediaPipe, RDKit

Deployment and Tools: FastAPI, ONNX Runtime, Streamlit, Git, GitHub, Docker, Supabase

AI AND MACHINE LEARNING PROJECTS

Indian Sign Language (ISL) Gesture-to-Text Recognition GitHub Ongoing

- Developing a recognition system for 50 ISL signs and one idle class using the AI4Bharat INCLUDE-50 dataset, comprising 943 videos.
- Built a landmark-based sequence pipeline using MediaPipe Pose and Hand Landmarkers; implemented a Conv1D-BiLSTM-Multi-Head Self-Attention model with approximately 0.88 million parameters.
- Developed inference components using FastAPI, WebSockets, and ONNX Runtime; 12 automated tests pass.

Technologies: Python, PyTorch, MediaPipe, OpenCV, FastAPI, WebSockets, ONNX Runtime

AromaScope – Molecular Odor Prediction Live Demo | GitHub 2026

- Developed a molecular odor-prediction system that accepts SMILES structures and predicts ranked odor descriptors across 113 odor classes.
- Engineered molecular representations using RDKit fingerprints and physicochemical properties for multi-label classification.
- Implemented scaffold-based model evaluation and applicability-domain checks to assess prediction reliability on molecular inputs.

Technologies: Python, RDKit, Scikit-learn, Pandas, NumPy, FastAPI, Docker

LifeOS 3.0 – AI-Powered Productivity and Learning Workspace Live Demo | GitHub 2026

- Built an integrated workspace combining academic planning, notes, tasks, focus sessions, habits, and exercise tracking.
- Implemented note-grounded AI explanations using retrieved passages, with citation checks to ensure responses are supported by source notes.
- Added selectable-text PDF extraction, private cloud file storage, optional authentication, cross-device synchronisation, and automated testing; 70 tests pass using simulated services.

Technologies: JavaScript, Supabase, PostgreSQL, Vercel, PDF.js, GitHub Actions

EDUCATION

Akal University, Talwandi Sabo, Punjab Aug. 2023 – Jun. 2027

B.Tech – Computer Science and Engineering (Data Science) **CGPA: 9.02/10**

RELEVANT COURSEWORK

Machine Learning, Artificial Intelligence, Deep Learning, Natural Language Processing, Probability and Statistics, Data Handling and Visualization, Data Structures and Algorithms, Database Management Systems

LEADERSHIP

Founding President – DataNinjas, Akal Tech Society Nov. 2024 – Dec. 2025

Helped grow a 75-member technical community; coordinated 10 volunteers and organised technical events, including a quiz, debate, and treasure hunt.

Student Learning Team Lead Feb. – May 2025

Led 30 students, monitored participation in daily English-learning sessions, and encouraged consistent engagement through follow-ups.

ACHIEVEMENTS

- Selected for the internal evaluation round of Smart India Hackathon (SIH) 2025.
- Secured **1st position** in a cybersecurity quiz among 10 participating teams.
- Qualified through two rounds of the CodeForge Club individual coding hackathon (2026).
- Led TechTrail '26, coordinating 5 members and organising an event involving 99 participants across 33 teams.